



1.1

Algorithm Design (2)

1.1.1 Algorithm Design: ISRO CSE 2023 | Question: 47



The asymptotic complexity of 4 functions f_1, f_2, f_3, f_4 are

1. $f_1(n) = 2^n$
2. $f_2(n) = n^{(3/2)}$
3. $f_3(n) = n \log n$
4. $f_4(n) = n^{(\log n)}$

Arrange them in increasing order of, complexity

- | | |
|-------------------------|-------------------------|
| A. f_2, f_3, f_4, f_1 | B. f_1, f_2, f_3, f_4 |
| C. f_3, f_2, f_4, f_1 | D. f_3, f_1, f_2, f_4 |

isro-cse-2023 asymptotic-notations time-complexity algorithm-design

Answer key

1.1.2 Algorithm Design: ISRO-DEC2017-44



A sorting technique is called stable if

- A. If it takes $O(n \log n)$ time.
- B. It uses divide and conquer technique.
- C. Relative order of occurrence of non-distinct elements is maintained.
- D. It takes $O(n)$ space.

isrodec2017 sorting algorithm-design data-structures

Answer key

1.2

Algorithm Design Techniques (5)

1.2.1 Algorithm Design Techniques: GATE CSE 1994 | Question: 1.19, ISRO2016-31



Algorithm design technique used in quicksort algorithm is?

- | | |
|------------------------|------------------|
| A. Dynamic programming | B. Backtracking |
| C. Divide and conquer | D. Greedy method |

gate1994 algorithms algorithm-design-techniques quick-sort easy isro2016

Answer key

1.2.2 Algorithm Design Techniques: GATE CSE 1998 | Question: 1.21, ISRO2008-16



Which one of the following algorithm design techniques is used in finding all pairs of shortest distances in a graph?

- | | |
|------------------------|-----------------------|
| A. Dynamic programming | B. Backtracking |
| C. Greedy | D. Divide and Conquer |

gate1998 algorithms algorithm-design-techniques easy isro2008

Answer key

1.2.3 Algorithm Design Techniques: ISRO CSE 2007 | Question: 55



Selection sort algorithm design technique is an example of

- A. Greedy method
- B. Divide-and-conquer
- C. Dynamic Programming
- D. Backtracking

isro2007 algorithms algorithm-design-techniques easy

Answer key

1.2.4 Algorithm Design Techniques: ISRO CSE 2011 | Question: 66



Which of the following algorithm design technique is used in merge sort?

- A. Greedy method
- B. Backtracking
- C. Dynamic programming
- D. Divide and Conquer

isro2011 algorithms algorithm-design-techniques merge-sort

Answer key

1.2.5 Algorithm Design Techniques: ISRO CSE 2018 | Question: 16



The following paradigm can be used to find the solution of the problem in minimum time:

Given a set of non-negative integer and a value K , determine if there is a subset of the given set with sum equal to K :

- a. Divide and Conquer
- b. Dynamic Programming
- c. Greedy Algorithm
- d. Branch and Bound

isro2018 algorithm-design-techniques algorithms

Answer key

1.3 Asymptotic Notations (2)

1.3.1 Asymptotic Notations: ISRO-DEC2017-18



Consider the recurrence equation

$$T(n) = \begin{cases} 2T(n-1), & \text{if } n > 0 \\ 1, & \text{otherwise} \end{cases}$$

Then $T(n)$ is (in *big O* order)

- A. $O(n)$
- B. $O(2^n)$
- C. $O(1)$
- D. $O(\log n)$

isrodec2017 recurrence-relation asymptotic-notations

Answer key

1.3.2 Asymptotic Notations: ISRO-DEC2017-19



```
//Consider the program
void function(int n)
{
    int i,j,count=0;
    for(i=n/2;i<=n;i++)
        for(j=1;j<=n;j=j*2)
            count++;
}
```

The complexity of the program is

- A. $O(\log n)$
- B. $O(n^2)$
- C. $O(n^2 \log n)$
- D. $O(n \log n)$

Answer key **1.4 Binary Search (2)****1.4.1 Binary Search: ISRO CSE 2007 | Question: 59**

The time taken by binary search algorithm to search a key in a sorted array of n elements is

- A. $O(\log_2 n)$ B. $O(n)$
C. $O(n \log_2 n)$ D. $O(n^2)$

isro2007 algorithms binary-search time-complexity

Answer key **1.4.2 Binary Search: ISRO CSE 2014 | Question: 28**

Suppose there are 11 items in sorted order in an array. How many searches are required on the average, if binary search is employed and all searches are successful in finding the item?

- A. 3.00
B. 3.46
C. 2.81
D. 3.33

algorithms binary-search isro2014

Answer key **1.5 Decision Table (1)****1.5.1 Decision Table: ISRO CSE 2007 | Question: 69**

A rule in a limited entry decision table is a

- A. row of the table consisting of condition entries
B. row of the table consisting of action entries
C. column of the table consisting of condition entries and the corresponding action entries
D. columns of the table consisting of conditions of the stub

isro2007 algorithms decision-table

Answer key **1.6 Dijkstras Algorithm (2)****1.6.1 Dijkstras Algorithm: ISRO CSE 2007 | Question: 80**

Dijkstra's algorithm is used to

- A. Create LSAs B. Flood an internet with information
C. Calculate the routing tables D. Create a link state database

isro2007 algorithms graph-algorithms shortest-path dijkstras-algorithm

Answer key **1.6.2 Dijkstras Algorithm: ISRO CSE 2023 | Question: 20**

Match the following:

(A) Floyd Warshall	(i) shortest path between two vertices
(B) Dijkstra	(ii) single source shortest path
(C) Kruskal's	(iii) Minimum spanning tree
(D) Bellman ford	(iv) solving all pair shortest path

A. (A)-(iii), (B)-(ii), (C)-(iv), (D)-(i)

C. (A)-(i), (B)-(ii), (C)-(iii), (D)-(iv)

B. (A)-(iv), (B)-(i), (C)-(iii), (D)-(ii)

D. (A)-(i), (B)-(ii), (C)-(iii), (D)-(iv)

isro-cse-2023 minimum-spanning-tree dijkstras-algorithm

Answer key

1.7 Dynamic Programming (1)

1.7.1 Dynamic Programming: ISRO-DEC2017-20



Match the following and choose the correct answer for the order A, B, C, D

A. Strassen matrix multiplication	p. Decrease and Conquer
B. Insertion sort	q. Dynamic Programming
C. Guassian Elimination	r. Divide and Conquer
D. Floyd shortest path algorithm	s. Transform and Conquer

A. a-r, b-s, c-p, d-q

C. a-q, b-s, c-p, d-r

B. a-r, b-p, c-s, d-q

D. a-s, b-p, c-q, d-r

isrodec2017 algorithms algorithm-design matrix dynamic-programming

Answer key

1.8 Expression Evaluation (1)

1.8.1 Expression Evaluation: ISRO CSE 2023 | Question: 43



The minimum number of arithmetic operations required to evaluate the expression $f(z) = 8x^3 + 3z + 12$

A. 4

B. 5

C. 3

D. 6

isro-cse-2023 arithmetic-series algorithm-design expression-evaluation

Answer key

1.9 Graph Algorithms (3)

1.9.1 Graph Algorithms: ISRO CSE 2017 | Question: 17



Which of the following data structure is useful in traversing a given graph by breadth first search?

A. Stack

B. Queue

C. List

D. None of the above

isro2017 data-structures graph-algorithms breadth-first-search easy

Answer key

1.9.2 Graph Algorithms: ISRO CSE 2017 | Question: 76



Which of the following algorithms solves the all pair shortest path problem?

- A. Prim's algorithm
- B. Dijkstra's algorithm
- C. Bellman ford algorithm
- D. Floyd warshalls algorithm

isro2017 algorithms graph-algorithms

Answer key

1.9.3 Graph Algorithms: ISRO CSE 2018 | Question: 33



Which of the following is application of Breath First Search on the graph?

- A. Finding diameter of the graph
- B. Finding bipartite graph
- C. Both (a) and (b)
- D. None of the above

isro2018 graph-algorithms breadth-first-search algorithms

Answer key

1.10 Hashing (3)

1.10.1 Hashing: GATE CSE 1989 | Question: 1-vii, ISRO2015-14



A hash table with ten buckets with one slot per bucket is shown in the following figure. The symbols $S1$ to $S7$ initially entered using a hashing function with linear probing. The maximum number of comparisons needed in searching an item that is not present is

0	S7
1	S1
2	
3	S4
4	S2
5	
6	S5
7	
8	S6
9	S3

- A. 4
- B. 5
- C. 6
- D. 3

hashing isro2015 gate1989 algorithms normal

Answer key

1.10.2 Hashing: ISRO CSE 2020 | Question: 24



In linear hashing, if blocking factor bfr , loading factor i and file buckets N are known, the number of records will be

- A. $cr = i + bfr + N$
- B. $r = i - bfr - N$
- C. $r = i + bfr - N$
- D. $r = i * bfr * N$

isro-2020 algorithms hashing normal

Answer key

1.10.3 Hashing: ISRO-DEC2017-50



The characters of the string K R P C S N Y T J M are inserted into a hash table of the size of size 10 using a hash function

$$h(x) = (ord(x) - ord(A) + 1) \bmod 10$$

If linear probing is used to resolve collisions, then the following insertion causes the collision

A. *Y*B. *C*C. *M*D. *P*

isrodec2017 hashing easy

Answer key

1.11

Huffman Code (1)

1.11.1 Huffman Code: ISRO CSE 2020 | Question: 34



Huffman tree is constructed for the following data : $\{A, B, C, D, E\}$ with frequency $\{0.17, 0.11, 0.24, 0.33 \text{ and } 0.15\}$ respectively. 100 00 01101 is decoded as

A. *BACE*B. *CADE*C. *BAD*D. *CADD*

isro-2020 algorithms huffman-code normal

Answer key

1.12

Identify Function (6)

1.12.1 Identify Function: ISRO CSE 2007 | Question: 23



Study the following program

```
//precondition: x>=0
public void demo(int x)
{
    System.out.print(x % 10);
    if (x % 10 != 0)
    {
        demo(x/10);
    }
    System.out.print(x%10);
}
```

Which of the following is printed as a result of the call demo (1234)?

A. 1441

B. 3443

C. 12344321

D. 43211234

isro2007 algorithms identify-function

Answer key

1.12.2 Identify Function: ISRO CSE 2008 | Question: 69



What is the value of $F(4)$ using the following procedure:

```
function F(K : integer)
integer;
begin
if (k<3) then F:=k else F:=F(k-1)*F(k-2)+F(k-3)
end;
```

A. 5

B. 6

C. 7

D. 8

isro2008 algorithms recursion identify-function

Answer key

1.12.3 Identify Function: ISRO CSE 2011 | Question: 40



Consider the following pseudocode

```
x:=1;
i:=1;
while ( x <= 500)
```

```
begin
x:=2^x;
i:=i+1;
end
```

What is the value of i at the end of the pseudocode?

- A. 4 B. 5 C. 6 D. 7

isro2011 algorithms identify-function

Answer key 

1.12.4 Identify Function: ISRO CSE 2018 | Question: 12



An array A consists of n integers in locations $A[0], A[1], \dots, A[n-1]$. It is required to shift the elements of the array cyclically to the left by k places, where $1 \leq k \leq (n-1)$. An incomplete algorithm for doing this in linear time, without using another array is given bellow. Complete the algorithm by filling in the blanks.

```
min=n; i=0;
while( ) {
    temp= A[i]; j=i;
    while( ) {
        A[j]= ;
        j=(j+k) mod n;
        if(j<min) then
            min = j;
    }
    A[(n+i-k) mod n]= ;
    i= ;
}
```

- a. $i > \min$; $j! = (n+1) \bmod n$; $A[j+k]$ temp; $i+1$;
b. $i < \min$; $j! = (n+i) \bmod n$; $A[j+k]$ temp; $i+1$;
c. $i > \min$; $j! = (n+i+k) \bmod n$; $A[j+k]$ temp; $i+1$;
d. $i < \min$; $j! = (n+i-k) \bmod n$; $A[(j+k) \bmod n]$ temp; $i+1$;

isro2018 algorithms identify-function

Answer key 

1.12.5 Identify Function: ISRO CSE 2018 | Question: 7



Assume A and B are non-zero positive integers. The following code segment:

```
while(A!=B){
    if(A>B)
        A -= B;
    else
        B -= A;
}
cout<<A; // printing the value of A
```

- a. Computes the *LCM* of two numbers b. Divides the larger number by the smaller number
c. Computes the *GCD* of two numbers d. Finds the smaller of two numbers

isro2018 algorithms identify-function

Answer key 

1.12.6 Identify Function: ISRO CSE 2018 | Question: 72



Consider the following C code segment

```

int f(int x)
{
    if(x<1) return 1;
    else return (if(x-1)+g(x));
}
int g(int x)
{
    if(x<2) return 2;
    else return (if(x-1)+g(x/2));
}

```

Of the following, which best describes the growth of $f(x)$ as a function of x ?

- a. Linear b. Exponential c. Quadratic d. Cubic

isro2018 algorithms identify-function time-complexity

Answer key 

1.13 Linear Probing (3)

1.13.1 Linear Probing: ISRO CSE 2014 | Question: 64



Consider a 13 element hash table for which $f(\text{key}) = \text{key} \bmod 13$ is used with integer keys. Assuming linear probing is used for collision resolution, at which location would the key 103 be inserted, if the keys 661, 182, 24 and 103 are inserted in that order?

- A. 0 B. 1 C. 11 D. 12

isro2014 hashing linear-probing

Answer key 

1.13.2 Linear Probing: ISRO CSE 2016 | Question: 29



A Hash Function f defined as $f(\text{key}) = \text{key} \bmod 7$. With linear probing while inserting the keys 37, 38, 72, 48, 98, 11, 56 into a table indexed from 0, in which location key 11 will be stored (Count table index 0 as 0^{th} location)?

- A. 3 B. 4 C. 5 D. 6

hashing isro2016 linear-probing

Answer key 

1.13.3 Linear Probing: ISRO CSE 2018 | Question: 36



A hash table with 10 buckets with one slot per bucket is depicted here. The symbols, S_1 to S_7 are initially entered using a hashing function with linear probing. The maximum number of comparisons needed in searching an item that is not present is

Answer key 

1.17

Merge Sort (3)

1.17.1 Merge Sort: GATE CSE 1999 | Question: 1.14, ISRO2015-42



If one uses straight two-way merge sort algorithm to sort the following elements in ascending order:

20, 47, 15, 8, 9, 4, 40, 30, 12, 17

then the order of these elements after second pass of the algorithm is:

- A. 8, 9, 15, 20, 47, 4, 12, 17, 30, 40
- B. 8, 15, 20, 47, 4, 9, 30, 40, 12, 17
- C. 15, 20, 47, 4, 8, 9, 12, 30, 40, 17
- D. 4, 8, 9, 15, 20, 47, 12, 17, 30, 40

gate1999 algorithms merge-sort normal isro2015 sorting

Answer key 

1.17.2 Merge Sort: ISRO CSE 2007 | Question: 58



The average case and worst case complexities for Merge sort algorithm are

- | | |
|----------------------------|-----------------------------------|
| A. $O(n^2), O(n^2)$ | B. $O(n^2), O(n \log_2 n)$ |
| C. $O(n \log_2 n), O(n^2)$ | D. $O(n \log_2 n), O(n \log_2 n)$ |

isro2007 algorithms sorting merge-sort

Answer key 

1.17.3 Merge Sort: ISRO CSE 2018 | Question: 35



Given two sorted list of size m and n respectively. The number of comparisons needed the worst case by the merge sort algorithm will be:

- | | |
|---------------------------|---------------------------|
| A. $m \times n$ | B. maximum of m and n |
| C. minimum of m and n | D. $m + n - 1$ |

isro2018 algorithms merge-sort

Answer key 

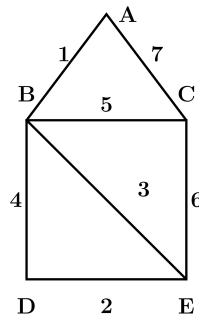
1.18

Minimum Spanning Tree (2)

1.18.1 Minimum Spanning Tree: ISRO CSE 2023 | Question: 19



Find the minimum spanning distance and the corresponding number of edges for the following graph



- A. 10,3 B. 11,4 C. 15,4 D. 28,7

isro-cse-2023 minimum-spanning-tree graph-algorithms algorithms

Answer key

1.18.2 Minimum Spanning Tree: ISRO2015-41



The number of spanning trees for a complete graph with seven vertices is

- A. 2^5 B. 7^5 C. 3^5 D. $2^{2 \times 5}$

isro2015 algorithms minimum-spanning-tree

Answer key

1.19

P NP NPC NPH (2)

1.19.1 P NP NPC NPH: GATE CSE 2004 | Question: 30, ISRO2017-10



The problem 3-SAT and 2-SAT are

- A. both in P B. both NP complete
C. NP-complete and in P respectively D. undecidable and NP complete respectively

gatecse-2004 algorithms p-np-npc-nph easy isro2017 out-of-gatecse-syllabus

Answer key

1.19.2 P NP NPC NPH: GATE CSE 2006 | Question: 16, ISRO-DEC2017-27



Let S be an NP-complete problem and Q and R be two other problems not known to be in NP. Q is polynomial time reducible to S and S is polynomial-time reducible to R. Which one of the following statements is true?

- A. R is NP-complete B. R is NP-hard
C. Q is NP-complete D. Q is NP-hard

gatecse-2006 algorithms p-np-npc-nph normal isrodec2017 out-of-gatecse-syllabus

Answer key

1.20

Quick Sort (2)

1.20.1 Quick Sort: ISRO-DEC2017-48



Quick-sort is run on 2 inputs shown below to sort in ascending order :

- A. $1, 2, 3, \dots, n$ B. $n, n-1, n-2, \dots, 1$

Let C_1 and C_2 be the number of comparisons made for **A** and **B** respectively. Then,

- A. $C1 > C2$
C. $C1 < C2$

- B. $C1 = C2$
D. Cannot say anything for arbitrary n

isrodec2017 sorting quick-sort algorithm-design asymptotic-notations

Answer key

1.20.2 Quick Sort: ISRO2015-12



A machine needs a minimum of 100 sec to sort 1000 names by quick sort. The minimum time needed to sort 100 names will be approximately

- A. 50.2 sec B. 6.7 sec C. 72.7 sec D. 11.2 sec

isro2015 algorithms quick-sort

Answer key

1.21 Recurrence Relation (5)

1.21.1 Recurrence Relation: GATE CSE 1994 | Question: 1.7, ISRO2017-14



The recurrence relation that arises in relation with the complexity of binary search is:

- A. $T(n) = 2T\left(\frac{n}{2}\right) + k$, k is a constant B. $T(n) = T\left(\frac{n}{2}\right) + k$, k is a constant
C. $T(n) = T\left(\frac{n}{2}\right) + \log n$ D. $T(n) = T\left(\frac{n}{2}\right) + n$

gate1994 algorithms recurrence-relation easy isro2017

Answer key

1.21.2 Recurrence Relation: GATE CSE 2004 | Question: 83, ISRO2015-40



The time complexity of the following C function is (assume $n > 0$)

```
int recursive (int n) {  
    if(n == 1)  
        return (1);  
    else  
        return (recursive (n-1) + recursive (n-1));  
}
```

- A. $O(n)$ B. $O(n \log n)$ C. $O(n^2)$ D. $O(2^n)$

gatecse-2004 algorithms recurrence-relation time-complexity normal isro2015

Answer key

1.21.3 Recurrence Relation: GATE CSE 2006 | Question: 51, ISRO2016-34



Consider the following recurrence:

$$T(n) = 2T(\sqrt{n}) + 1, T(1) = 1$$

Which one of the following is true?

- A. $T(n) = \Theta(\log \log n)$ B. $T(n) = \Theta(\log n)$
C. $T(n) = \Theta(\sqrt{n})$ D. $T(n) = \Theta(n)$

algorithms recurrence-relation isro2016 gatecse-2006

Answer key

1.21.4 Recurrence Relation: ISRO CSE 2011 | Question: 56



Let $T(n)$ be defined by $T(1) = 10$ and $T(n+1) = 2n + T(n)$ for all integers $n \geq 1$. Which of the following represents the order of growth of $T(n)$ as a function of n ?

- A. $O(n)$ B. $O(n \log n)$ C. $O(n^2)$ D. $O(n^3)$

isro2011 algorithms time-complexity recurrence-relation

Answer key

1.21.5 Recurrence Relation: ISRO CSE 2018 | Question: 37



The running time of an algorithm is given by:

$$T(n) = T(n-1) + T(n-2) - T(n-3), \text{ if } n > 3 \\ = n, \text{ otherwise}$$

Then what should be the relation between $T(1), T(2), T(3)$, so that the order of the algorithm is constant?

- a. $T(1) = T(2) = T(3)$ b. $T(1) + T(3) = 2T(2)$
c. $T(1) - T(3) = T(2)$ d. $T(1) + T(2) = T(3)$

isro2018 algorithms recurrence-relation time-complexity

Answer key

1.22

Recursion (1)

1.22.1 Recursion: ISRO-DEC2017-73



Consider the function

```
int fun(x: integer)
{
    if x>100 then fun=x-10;
    else
        fun=fun(fun(x+11));
}
```

For the input $x = 95$, the function will return

- A. 89 B. 90 C. 91 D. 92

isrodec2017 programming-in-c recursion

Answer key

1.23

Relations (1)

1.23.1 Relations: ISRO CSE 2017 | Question: 80



The time complexity of computing the transitive closure of a binary relation on a set of n elements is known to be

- a. $O(n \log n)$
b. $O(n^{3/2})$
c. $O(n^3)$
d. $O(n)$

Answer key 

1.24

Searching (2)

1.24.1 Searching: GATE CSE 1996 | Question: 2.13, ISRO2016-28



The average number of key comparisons required for a successful search for sequential search on n items is

- A. $\frac{n}{2}$ B. $\frac{n-1}{2}$ C. $\frac{n+1}{2}$ D. None of the above

gate1996 algorithms easy isro2016 searching

Answer key 

1.24.2 Searching: ISRO CSE 2011 | Question: 70



Number of comparisons required for an unsuccessful search of an element in a sequential search organized, fixed length, symbol table of length L is

- A. L B. $L/2$ C. $(L+1)/2$ D. $2L$

isro2011 algorithms searching

Answer key 

1.25

Sorting (10)

1.25.1 Sorting: GATE CSE 2006 | Question: 14, ISRO2011-14



Which one of the following in place sorting algorithms needs the minimum number of swaps?

- A. Quick sort B. Insertion sort C. Selection sort D. Heap sort

gatecse-2006 algorithms sorting easy isro2011

Answer key 

1.25.2 Sorting: ISRO CSE 2008 | Question: 77



How many comparisons are needed to sort an array of length 5 if a straight selection sort is used and array is already in the opposite order?

- A. 1 B. 10 C. 15 D. 20

isro2008 algorithms sorting selection-sort

Answer key 

1.25.3 Sorting: ISRO CSE 2014 | Question: 62



Consider the following sorting algorithms.

- I. Quicksort
- II. Heapsort
- III. Mergesort

Which of them perform in least time in the worst case?

- A. I and II only B. II and III only C. III only D. I, II and III

isro2014 algorithms sorting

Answer key 

1.25.4 Sorting: ISRO CSE 2017 | Question: 15



Which one of the following in-place sorting algorithms needs the minimum number of swaps?

- A. Insertion Sort B. Quick Sort C. Heap Sort D. Selection Sort

isro2017 algorithms sorting

Answer key 

1.25.5 Sorting: ISRO CSE 2017 | Question: 49



The number of swappings needed to sort the numbers 8, 22, 7, 9, 31, 5, 13 in ascending order using bubble sort is

- A. 11 B. 12 C. 13 D. 10

isro2017 algorithms sorting

Answer key 

1.25.6 Sorting: ISRO CSE 2018 | Question: 66



Of the following sorting algorithms, which has a running time that is least dependent on the initial ordering of the input?

- a. Merge Sort b. Insertion Sort c. Selection Sort d. Quick Sort

isro2018 algorithms sorting

Answer key 

1.25.7 Sorting: ISRO CSE 2020 | Question: 33



If an array A contains the items 10, 4, 7, 23, 67, 12 and 5 in that order, what will be the resultant array A after third pass of insertion sort?

- A. 67, 12, 10, 5, 4, 7, 23 B. 4, 7, 10, 23, 67, 12, 5
C. 4, 5, 7, 67, 10, 12, 23 D. 10, 7, 4, 67, 23, 12, 5

isro-2020 algorithms sorting normal

Answer key 

1.25.8 Sorting: ISRO CSE 2020 | Question: 65



Of the following sort algorithms, which has execution time that is least dependant on initial ordering of the input?

- A. Insertion sort B. Quick sort C. Merge sort D. Selection sort

isro-2020 algorithms sorting normal

Answer key 

1.25.9 Sorting: ISRO CSE 2023 | Question: 6



Worst case time complexity of heap sort for n elements?

- A. $O(n \log n)$ B. $O(\log n)$ C. $O(n^2)$ D. $O(n)$

isro-cse-2023 algorithms heap-sort sorting

Answer key 

1.25.10 Sorting: ISRO-2013-12



Which of the following sorting algorithms has the minimum running time complexity in the best and average case?

- A. Insertion sort, Quick sort
- B. Quick sort, Quick sort
- C. Quick sort, Insertion sort
- D. Insertion sort, Insertion sort

isro2013 sorting

Answer key

1.26

Symbol Table (1)

1.26.1 Symbol Table: ISRO CSE 2016 | Question: 39



Access time of the symbolic table will be logarithmic if it is implemented by

- A. Linear list
- B. Search tree
- C. Hash table
- D. Self organization list

isro2016 algorithms symbol-table

Answer key

1.27

Time Complexity (3)

1.27.1 Time Complexity: GATE CSE 2007 | Question: 15, ISRO2016-26



Consider the following segment of C-code:

```
int j, n;  
j = 1;  
while (j <= n)  
    j = j * 2;
```

The number of comparisons made in the execution of the loop for any $n > 0$ is:

- A. $\lceil \log_2 n \rceil + 1$
- B. n
- C. $\lfloor \log_2 n \rfloor$
- D. $\lfloor \log_2 n \rfloor + 1$

gatecse-2007 algorithms time-complexity normal isro2016

Answer key

1.27.2 Time Complexity: ISRO CSE 2014 | Question: 67



What is the time complexity for the following C module? Assume that $n > 0$.

```
int module(int n)  
{  
    if (n == 1)  
        return 1;  
    else  
        return (n + module(n-1));  
}
```

- A. $O(n)$
- B. $O(\log n)$
- C. $O(n^2)$
- D. $O(n!)$

isro2014 algorithms time-complexity

Answer key

1.27.3 Time Complexity: ISRO CSE 2020 | Question: 36



What is the complexity of the following code?

```
sum=0;
```

```
for(i=1;i<=n;i*=2)
for(j=1;j<=n;j++)
sum++;
```

Which of the following is not a valid string?

- A. $O(n^2)$ B. $O(n \log n)$ C. $O(n)$ D. $O(n \log n \log n)$

isro-2020 algorithms time-complexity normal

Answer key 

Answer Keys

1.1.1	C	1.1.2	C	1.2.1	C	1.2.2	A	1.2.3	X
1.2.4	D	1.2.5	B	1.3.1	B	1.3.2	D	1.4.1	A
1.4.2	A	1.5.1	C	1.6.1	C	1.6.2	B	1.7.1	B
1.8.1	D	1.9.1	B	1.9.2	D	1.9.3	C	1.10.1	B
1.10.2	D	1.10.3	X	1.11.1	A	1.12.1	D	1.12.2	A
1.12.3	B	1.12.4	D	1.12.5	C	1.12.6	B	1.13.1	B
1.13.2	C	1.13.3	B	1.14.1	B	1.15.1	B	1.16.1	B
1.17.1	B	1.17.2	D	1.17.3	D	1.18.1	B	1.18.2	B
1.19.1	C	1.19.2	B	1.20.1	B	1.20.2	B	1.21.1	B
1.21.2	D	1.21.3	B	1.21.4	C	1.21.5	X	1.22.1	C
1.23.1	C	1.24.1	C	1.24.2	A	1.25.1	C	1.25.2	B
1.25.3	B	1.25.4	D	1.25.5	D	1.25.6	A	1.25.7	B
1.25.8	C	1.25.9	A	1.25.10	A	1.26.1	B	1.27.1	D
1.27.2	A	1.27.3	X						